SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Artificial Intelligence

Practical Rule Mining

Aim : Association Rule Mining

- Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.
- Generate association rules from the frequent itemsets and calculate their support and confidence.
- Interpret and analyze the discovered association rules.

CODE:

T074 Ai Pract Rule Mining.py - C:/Users/PRIYANKA/Downloads/T074 Ai Pract Rule Mining.py (3.13.1)

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```python
import pandas as pd
import matplotlib.pyplot as plt
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules

df = pd.read_csv(r"C:\Users\PRIYANKA\Downloads\gta_data_batch_1.csv")

columns = [
    "vehicle_class",
    "manufacturer",
    "acquisition",
    "storage_location",
    "delivery_method",
    "modifications",
    "race_availability",
    "bulletproof"
]

transactions = []

for _, row in df.iterrows():
    items = []

    for col in columns:
        if col in df.columns and pd.notna(row[col]):
            items.append(col + "=" + str(row[col]))

    if "features" in df.columns and pd.notna(row["features"]):
        features = str(row["features"]).split(",")
        for feature in features:
            feature = feature.strip()
            if feature:
                items.append("Feature=" + feature)

    transactions.append(items)

te = TransactionEncoder()

transaction_data = te.fit(transactions).transform(transactions)

transaction_df = pd.DataFrame(
    transaction_data,
    columns=te.columns_
)

frequent_itemsets = apriori(
    transaction_df,
```

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```python
frequent_itemsets = apriori(
    transaction_df,
    min_support=0.08,
    use_colnames=True
)

frequent_itemsets = frequent_itemsets.sort_values(
    by="support",
    ascending=False
)

print("Frequent Itemsets:")
print(frequent_itemsets.head(20))

rules = association_rules(
    frequent_itemsets,
    metric="confidence",
    min_threshold=0.5
)

rules = rules.sort_values(
    by=["confidence", "lift"],
    ascending=False
)

print("\nAssociation Rules:")
print(
    rules[
        [
            "antecedents",
            "consequents",
            "support",
            "confidence",
            "lift"
        ]
    ].head(20)
)

single_itemsets = frequent_itemsets[
    frequent_itemsets["itemsets"].apply(lambda x: len(x) == 1)
].head(10).copy()

single_itemsets["itemsets"] = single_itemsets["itemsets"].apply(
    lambda x: list(x)[0]
)

plt.figure(figsize=(10, 6))
```

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```python
    lambda x: list(x)[0]
)

plt.figure(figsize=(10, 6))

plt.barh(
    single_itemsets["itemsets"],
    single_itemsets["support"]
)

plt.title("Top 10 Frequent Items")
plt.xlabel("Support")
plt.ylabel("Items")
plt.xlim(0, 1)
plt.tight_layout()
plt.show()

top_rules = rules.head(10).copy()

top_rules["rule"] = top_rules.apply(
    lambda x: (
        " + ".join(list(x["antecedents"]))
        + " -> "
        + " + ".join(list(x["consequents"]))
    ),
    axis=1
)

top_rules = top_rules.sort_values(
    by="confidence",
    ascending=True
)

plt.figure(figsize=(12, 7))

plt.barh(
    top_rules["rule"],
    top_rules["confidence"]
)

plt.title("Top 10 Association Rules")
plt.xlabel("Confidence")
plt.ylabel("Rules")
plt.xlim(0, 1)
plt.tight_layout()
plt.show()
```

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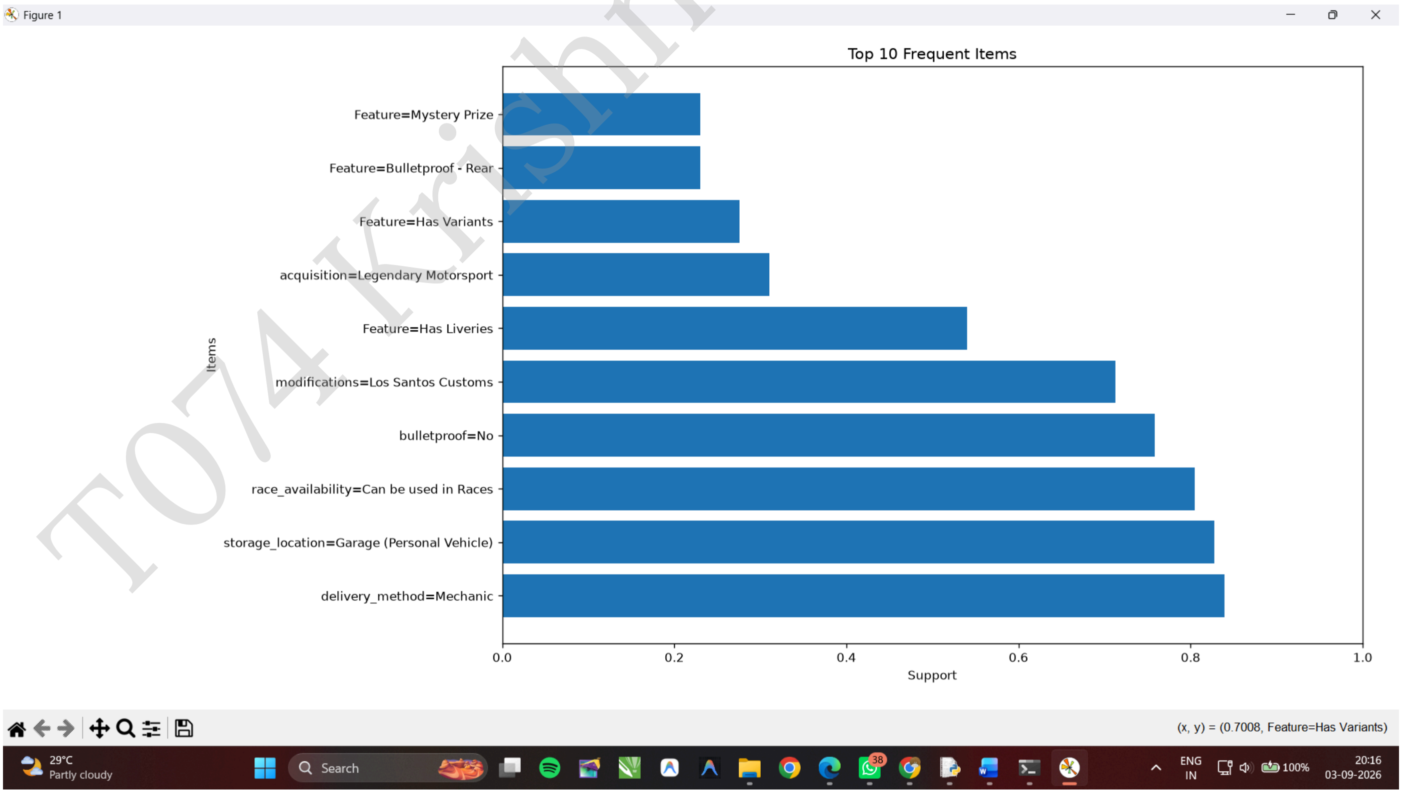
OUTPUT:

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
====== RESTART: C:/Users/PRIYANKA/Downloads/T074 Ai Pract Rule Mining.py ======
Frequent Itemsets:
      support                                           itemsets
19   0.839080              frozenset({delivery_method=Mechanic})
147  0.827586  frozenset({storage_location=Garage (Personal V...
26   0.827586  frozenset({storage_location=Garage (Personal V...
24   0.804598  frozenset({race_availability=Can be used in Ra...
146  0.781609  frozenset({race_availability=Can be used in Ra...
167  0.781609  frozenset({race_availability=Can be used in Ra...
435  0.781609  frozenset({race_availability=Can be used in Ra...
18   0.758621                    frozenset({bulletproof=No})
132  0.724138  frozenset({bulletproof=No, delivery_method=Mec...
395  0.724138  frozenset({storage_location=Garage (Personal V...
137  0.724138  frozenset({storage_location=Garage (Personal V...
23   0.712644       frozenset({modifications=Los Santos Customs})
742  0.712644  frozenset({race_availability=Can be used in Ra...
145  0.712644  frozenset({modifications=Los Santos Customs, d...
428  0.712644  frozenset({modifications=Los Santos Customs, s...
160  0.712644  frozenset({modifications=Los Santos Customs, s...
136  0.712644  frozenset({race_availability=Can be used in Ra...
394  0.712644  frozenset({race_availability=Can be used in Ra...
412  0.712644  frozenset({race_availability=Can be used in Ra...
159  0.701149  frozenset({modifications=Los Santos Customs, r...

Association Rules:
                                             antecedents  ...       lift
5399  frozenset({race_availability=Can be used in Ra...  ...  12.428571
5403  frozenset({modifications=Benny's Original Moto...  ...  12.428571
5405  frozenset({modifications=Benny's Original Moto...  ...  12.428571
5408  frozenset({modifications=Benny's Original Moto...  ...  12.428571
5412  frozenset({modifications=Benny's Original Moto...  ...  12.428571
5415  frozenset({race_availability=Can be used in Ra...  ...  12.428571
5416  frozenset({race_availability=Can be used in Ra...  ...  12.428571
5417  frozenset({race_availability=Can be used in Ra...  ...  12.428571
5418  frozenset({race_availability=Can be used in Ra...  ...  12.428571
5419  frozenset({bulletproof=No, acquisition=Benny's...  ...  12.428571
5420  frozenset({race_availability=Can be used in Ra...  ...  12.428571
5422  frozenset({modifications=Benny's Original Moto...  ...  12.428571
5423  frozenset({modifications=Benny's Original Moto...  ...  12.428571
5424  frozenset({modifications=Benny's Original Moto...  ...  12.428571
5426  frozenset({acquisition=Benny's Original Motor ...  ...  12.428571
5427  frozenset({acquisition=Benny's Original Motor ...  ...  12.428571
5428  frozenset({modifications=Benny's Original Moto...  ...  12.428571
5430  frozenset({modifications=Benny's Original Moto...  ...  12.428571
```



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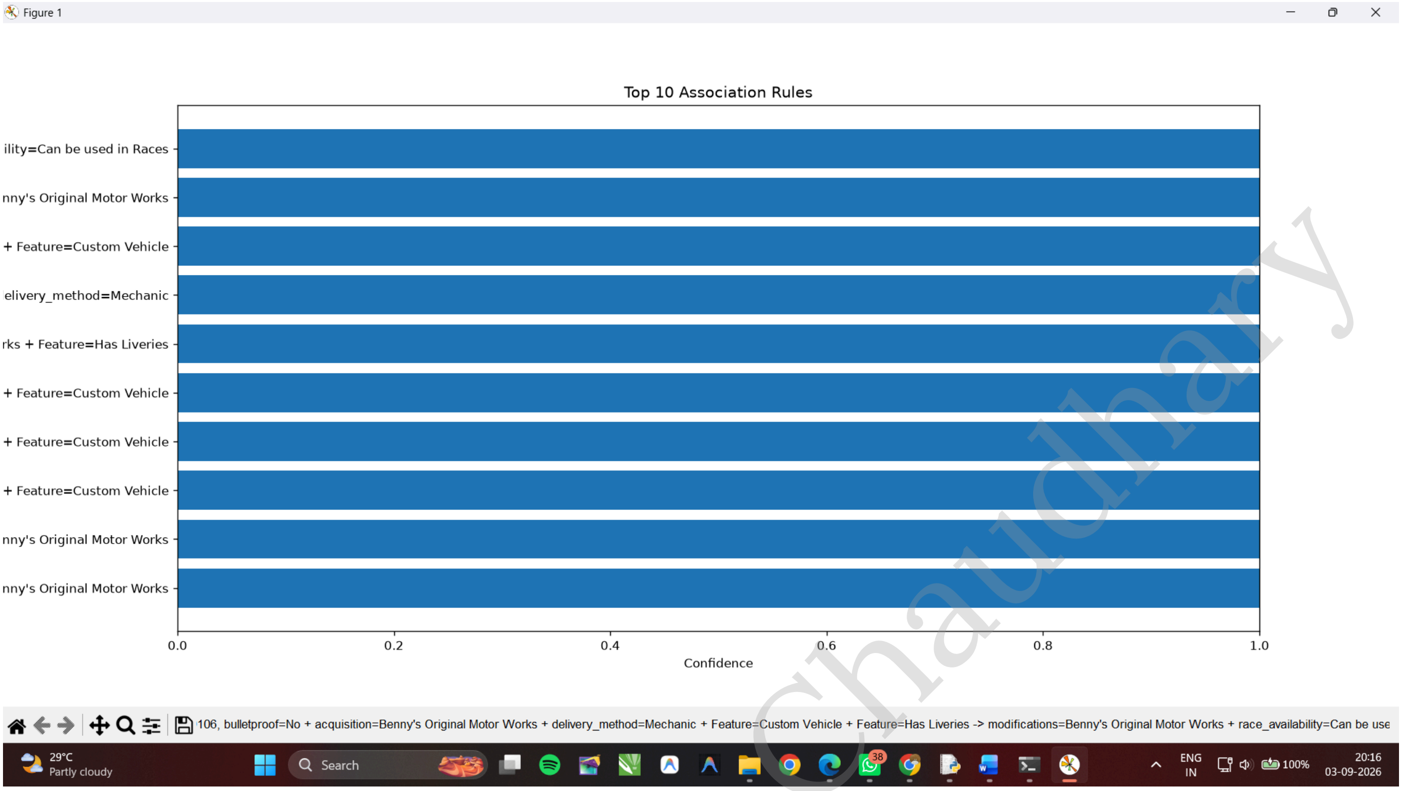

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